Using Mobile Phone Data in Economics and Business Research

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Stillman Research Day
April 30, 2026

Motivation

- Mobile phones are ubiquitous worldwide, 112 subscriptions per 100 people (World Bank, 2025)
- They create massive amounts of passive data with use in behavioral settings (location, usage, mobile money)
- Phones themselves can alter economic outcomes (network effects, information channels, etc.)

Research seed: Explore the ways economic and business researchers use phone (or related) data

As a measurement tool:

1. Inferring migration patterns from call detail records
2. Predicting poverty and wealth
3. Managing the spread of disease

As a variable of interest:

1. How does mobile coverage affect violence in an insurgency?
2. Does consumer access to Google Maps affect firm location choice?
3. Will a country's level of mobile internet access affect its rate of emigration?

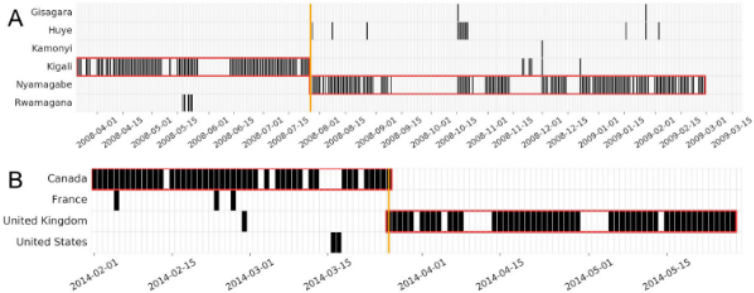

Fig 4. Trajectories of two migration events detected by our approach. Each row is a different district, each column is a day; cells are black if the individual makes or receives a phone call from that district on that day. Red boxes indicate location segments. Orange line marks the inferred date of migration. (A) Long-term migration in Rwanda (migrated from Kigali to Nyamagabe on 2008-07-27). (B) International migration (migrated from Canada to United Kingdom on 2014-03-26).

https://doi.org/10.1371/journal.pone.0239408.g004

Figure: Inferring migration events
Chi et al., 2020

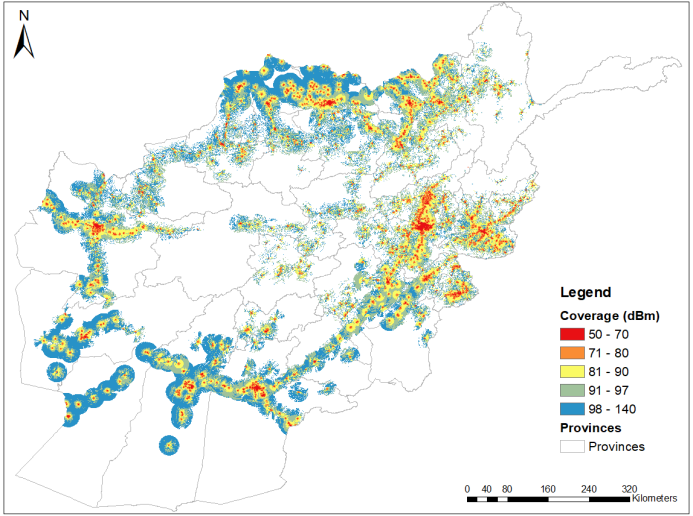

Figure: Irregular Terrain Model
Ellis et al., 2026

Data

- Some data (e.g. GPS) is very difficult to get and prohibitively expensive
- Other types (CDR) are comparatively less risky for mobile network operators to share with academic groups
- Measures of coverage have both paid and open-source options (GSMA, OpenCellID)

Goal for today

An idea-generating "sprint," after which we'll vote for a project worth pursuing

- Start broadly: what is a topic you're interested in researching?
- How can it be measured? (it's okay if phone data isn't the main source)
- Can your question be addressed (fully or partially) by either:
  1. the positioning and movement of individuals?
  2. the amount of coverage in an area?

- The examples to this point are not exhaustive. Applications are everywhere. Be creative and follow your intution!